NETWORK RESILIENCE ANALYZER

A Python study of critical devices and communication network failure strategies

COURSE PROJECT	**ITW III ? Batch 2**
TECHNOLOGY	Python ? NetworkX ? Matplotlib
LEVEL / DURATION	Intermediate ? one month

A reproducible notebook-based project. The notebook builds its sample topology, runs the analysis, and displays the generated figures.

1. Problem statement

Communication networks depend on connected devices and links. A network may have alternate routes, but a small number of devices can still connect otherwise separate areas. This project models a communication network, identifies structurally important devices, and evaluates how device failures affect connectivity. The central question is whether targeted removal of important devices reduces network reachability faster than random failures.

2. Objectives

? Build and visualize a sample topology of routers, switches, servers, an access point, and client devices.

? Measure connectivity, density, average shortest-path length, degree centrality, and betweenness centrality.

? Compare random failures with targeted removal using pairwise reachability as the outcome.

? Interpret the findings and explain the assumptions and limitations.

3. Tools and methodology

Python runs the implementation; **NetworkX** models and measures the graph; **Matplotlib** creates the monochrome network diagram and charts. No external dataset is needed. Devices are nodes and links are undirected edges.

Stage	Method
Baseline	Compute graph connectivity, density, mean shortest-path length, and centrality scores.
Failure cases	For removal levels from 0% to 80%, sample random node removals 200 times and remove nodes in descending origi
Outcome	Divide the number of connected unordered pairs remaining by the number of all unordered pairs in the original netwo

4. Implementation plan

Week	Planned work
1	Define scope, build topology, check assumptions and graph validity.
2	Implement graph visualization and baseline metrics.
3	Implement failure strategies and resilience comparison plots.
4	Interpret results, polish notebook and report, prepare demonstration.

5. Results

The notebook was executed successfully with a fixed random seed. Its example network contains **14 devices and 17 links**; it is connected, has density **0.187**, and an average shortest-path length of **2.58**. The highest betweenness scores are Switch_2 (0.498), Router_B (0.459), and Switch_1 (0.402), indicating that these devices lie on many shortest routes in this topology.

Devices removed	Random failures?	Targeted removal?
0%	1.000	1.000
10%	0.779	0.604
20%	0.440	0.055
30%	0.316	0.033
40%	0.135	0.011
50%	0.093	0.011
60%	0.059	0.011
70%	0.018	0.011
80%	0.009	0.011

? Mean across 200 random samples at each removal level. ? Nodes removed in descending betweenness order computed on the original graph. Values are the reachable fraction of all original device pairs. At 20% removal, random failure retains 0.440 of pairs on average while targeted removal retains 0.055. This example illustrates how a few central devices can strongly affect reachability. The small difference at 80% is due to the surviving node/pair count and rounding; it does not change the overall pattern.

6. Discussion and conclusion

The results show that centrality can help identify devices whose removal may fragment a network. In this example, targeted failures reduce pairwise reachability more quickly than random failures at most of the measured levels. The conclusion is specific to this hand-built topology and simulation design, rather than a general claim about all real networks. The project demonstrates a compact, reproducible use of graph analysis to discuss network resilience and the value of alternate paths.

7. Limitations and future work

The model uses undirected links, treats all device pairs equally, and removes devices instantly. It omits link failures, capacity, traffic demand, latency, routing protocols, and recovery. Targeted removal uses centrality measured before the first failure. Future improvements could add weighted links, link-failure experiments, adaptive re-ranking after each removal, or user-provided topologies.

8. Demonstration outline

? Explain how nodes and links map to the example network.

? Compare degree centrality with betweenness centrality.

? Run and interpret the random-versus-targeted reachability curve.

? Discuss one limitation and propose a redundancy improvement.

Submission files: network_resilience_analyzer.ipynb (implementation, computed output and plots) and this project report.